Bank Customer Churn Analysis

Business Analytics Project Report

Prepared by	Rahul Panigrahi
Dataset	Bank_Churn.csv
Total Records	10,000 customers
Tools Used	Python · Power BI
Analysis Type	Exploratory + Segment Analysis

1. Executive Summary

This report analyses 10,000 bank customers to understand what is driving a 20.37% churn rate — and what the bank can do about it. The analysis is structured in three parts: data exploration to understand broad patterns, deep segment analysis to find root causes, and recommendations for the retention team.

The core finding is straightforward. The bank is not losing random customers. It is losing a specific profile: older customers aged 41-60, in Germany, who are inactive and hold 3-4 bank products. This group churns at 94-95% — nearly five times the overall rate. More importantly, churned customers hold higher account balances than those who stay — meaning the bank is losing its most financially valuable customers.

Key Metric	Value
Total customers	10,000
Customers who left (churned)	2,037 (20.37%)
Customers retained	7,963 (79.63%)
Country with highest churn	Germany — 32.44%
Age group with highest churn	51 to 60 — 56.21%
Most dangerous product segment	3 to 4 products — 82 to 100%
Avg balance of churned customers	91,108 vs 72,745 for retained
Highest risk segment identified	German inactive 41-60 yr olds with 3-4 products — 94.87%

2. Dataset Overview

2.1 About the Data

The dataset contains records of 10,000 customers from a European bank operating across France, Germany, and Spain. Each row represents one customer with 13 attributes covering their demographics, financial behaviour, and whether they exited the bank.

Column	Description
CustomerId	Unique customer identifier — not used in analysis
Surname	Customer name — not used in analysis
CreditScore	Customer credit score ranging from 350 to 850
Geography	Country — France, Germany, or Spain
Gender	Male or Female
Age	Customer age in years
Tenure	Number of years the customer has been with the bank
Balance	Account balance
NumOfProducts	Number of bank products the customer holds (1 to 4)
HasCrCard	Whether the customer has a credit card — 1 yes, 0 no
IsActiveMember	Whether the customer is active — 1 yes, 0 no
EstimatedSalary	Estimated annual salary
Exited	Whether the customer left — 1 churned, 0 retained (Target)

2.2 Data Quality

- Zero null values across all 13 columns — no imputation needed
- No duplicate rows detected
- All data types are correctly formatted
- Two new columns were created during analysis: Age_Group and Balance_Segment

3. Data Exploration

3.1 Overall Churn Rate

Status	Count	Percentage
Retained	7,963	79.63%
Churned	2,037	20.37%
Total	10,000	100%

Note: *Industry benchmark for retail banking churn is typically 10 to 15%. At 20.37%, this bank is operating at nearly double the acceptable rate.*

3.2 Churn by Geography

Country	Customers	Churned	Churn Rate
Germany	2,509	814	32.44%
Spain	2,477	413	16.67%
France	5,014	810	16.15%

Germany churns at 32.44% — double the rate of France and Spain. Germany holds 25% of the customer base but accounts for 40% of total churn. This gap is too large to be explained by random variation.

3.3 Churn by Age Group

Age Group	Customers	Churned	Churn Rate
51 to 60	797	448	56.21%
41 to 50	2,320	788	33.97%
60+	464	115	24.78%
31 to 40	4,451	538	12.09%
Under 30	1,968	148	7.52%

Churn rises sharply from age 41 onwards. The 51-60 group loses more than half its customers. These customers are typically at peak earning years with the highest account balances — their departure carries the greatest financial impact.

3.4 Active vs Inactive Members

Status	Customers	Churned	Churn Rate
Inactive	4,849	1,302	26.85%
Active	5,151	735	14.27%

Inactive members churn at nearly double the rate of active members. A customer who stops using their account is signalling their intention to leave — typically 2 to 3 months before formally exiting. Activity status is a leading indicator, not a lagging one.

3.5 Churn by Number of Products

Products Held	Customers	Churned	Churn Rate
1 product	5,084	1,409	27.71%
2 products	4,590	348	7.58%
3 products	266	220	82.71%
4 products	60	60	100.00%

Two products is the retention sweet spot at 7.58% churn. Single product customers lack loyalty and churn at 27.71%. Most alarmingly, every single customer holding 4 products left the bank. Customers holding 3 to 4 products are not more loyal — they are over-sold. They acquired products they did not need, became disengaged, and left.

3.6 Balance — Churned vs Retained

Status	Average Balance	Median Balance
Churned	91,108	109,349
Retained	72,745	92,073
Difference	+18,363 higher	+17,276 higher

Churned customers hold an average balance 18,363 higher than retained customers. The bank is not losing low-value customers — it is losing its most financially significant ones. The revenue impact of each churned customer is above average.

3.7 Churn by Gender

Gender	Customers	Churned	Churn Rate
Female	4,543	1,139	25.07%
Male	5,457	898	16.46%
Gap	—	—	8.61% difference

Female customers churn 8.61% more than male customers. This gap is consistent enough to suggest that the bank's products, communication, or service experience is not resonating equally with female customers.

3.8 Balance by Segment

Balance Segment	Customers	Churned	Churn Rate
Low (0 to 50K)	75	26	34.67%
Mid (50K to 100K)	1,509	300	19.88%
High (100K to 150K)	3,830	987	25.77%
Premium (150K+)	969	224	23.12%

The High balance segment (100K to 150K) has the largest absolute churn count at 987 customers. Combined with the earlier finding that churned customers hold higher balances overall, this confirms that mid-to-high balance customers represent the greatest retention risk and the greatest financial priority.

3.9 Variables With Negligible Impact on Churn

Variable	Churned Avg	Retained Avg	Difference	Conclusion
Credit Score	645	652	7 points	Not meaningful
Tenure	4.93 years	5.03 years	0.1 years	Not meaningful
Estimated Salary	101,466	99,738	1,728	Not meaningful
Has Credit Card	20.18%	20.81%	0.63%	Not meaningful

Credit score, tenure, salary, and credit card ownership show no meaningful difference between customers who churned and those who stayed. Retention strategies targeting these variables would be misaligned.

4. Deep Analysis — Churn Risk Funnels

Two analytical funnels were constructed using a Segment Decomposition approach. The method starts with a high-level finding from the exploration phase and progressively narrows by adding one variable at a time. The goal is to identify the most precise high-risk customer profile — not just who churns, but the specific combination of factors that drives extreme churn.

4.1 Funnel 1 — Germany Deep Dive

Starting point: Germany churns at 32.44% — double France and Spain

Before narrowing the funnel, the Germany data was broken down by each variable individually to understand what is different about German customers specifically.

Variable	Germany Rate	Overall Rate	Difference
Age 51-60	69.55%	56.21%	+13.34%
Age 41-50	48.82%	33.97%	+14.85%
Female customers	37.55%	25.07%	+12.48%
Male customers	27.81%	16.46%	+11.35%
Inactive members	41.08%	26.85%	+14.23%
3 to 4 products	89 to 100%	82 to 100%	Consistent — global issue

Every single variable is worse in Germany than the overall average. Germany does not have one isolated problem — it has an amplified version of every problem the bank faces.

Narrowing the funnel

Segment	Customers	Churned	Churn Rate	vs Overall
Overall	10,000	2,037	20.37%	Baseline
Germany	2,509	814	32.44%	1.6x higher
Germany + Age 41 to 60	921	500	54.29%	2.7x higher
Germany + Inactive + Age 41 to 60	501	323	64.47%	3.2x higher
Germany + Inactive + Age 41-60 + 3-4 Products	39	37	94.87%	4.7x higher

An inactive German customer aged 41 to 60 holding 3 to 4 bank products has a 94.87% churn rate. Out of 39 customers in this profile, 37 left the bank. The bank cross-sold multiple products to its most valuable demographic in Germany. Those customers became disengaged and left — taking all products and all revenue with them.

4.2 Funnel 2 — Age Group Deep Dive

Starting point: Age 41-60 has elevated churn globally — is this a Germany issue or everywhere?

Funnel 2 tests whether the age-driven churn pattern holds across all three countries or is specific to Germany. The same narrowing approach is applied globally.

Narrowing the funnel

Segment	Customers	Churned	Churn Rate	vs Overall
Overall	10,000	2,037	20.37%	Baseline
Age 41 to 60	3,117	1,236	39.65%	1.9x higher
Inactive + Age 41 to 60	1,562	798	51.09%	2.5x higher
Inactive + Age 41-60 + 3-4 Products	106	101	95.28%	4.7x higher

The result is almost identical to Funnel 1 at 95.28%. This confirms that the over-selling problem is not limited to Germany — it exists globally. Inactive 41 to 60 year olds with 3 to 4 products churn at 95% regardless of country.

4.3 What Both Funnels Tell Us Together

Funnel	Final Segment	Churn Rate	Multiplier
Funnel 1 — Germany	German + Inactive + 41-60 + 3-4 Products	94.87%	4.7x
Funnel 2 — Age	Inactive + 41-60 + 3-4 Products (all countries)	95.28%	4.7x

Both funnels independently arrive at the same conclusion. The combination of inactivity, age 41 to 60, and 3 to 4 products produces a near-certain churner regardless of geography. Germany amplifies this problem but the root cause — aggressive cross-selling to disengaged older customers — is a bank-wide issue.

5. Recommendations

Recommendations are grouped into three priority tiers. Priority 1 addresses the most urgent problems with the largest financial impact. Priority 2 covers structural fixes that prevent future churn. Priority 3 covers strategic changes that build long-term retention capability.

Priority 1 — Immediate Action

#	What to Do	Who to Target	Why
1	Audit all customers with 3 to 4 products — review product relevance and fee burden	326 customers: 266 with 3 products, 60 with 4	These customers churn at 82-100%. Intervention now may recover some before they exit
2	Launch Germany-specific retention programme with dedicated relationship managers	814 already churned, 1,695 still retained in Germany	Germany churn at 32% — double other markets. Needs focused attention
3	Re-engage all inactive customers aged 41 to 60 with personalised outreach	1,562 inactive customers in the 41-60 age range	This group churns at 51%. A targeted call or offer could recover a meaningful portion

Priority 2 — Short Term (1 to 4 Weeks)

#	What to Do	Who to Target	Why
4	Stop cross-selling beyond 2 products without a clear customer need assessment	All new and existing customers	3-4 product customers churn at 82-100%. Over-selling destroys rather than builds loyalty
5	Cross-sell one relevant second product to all single-product customers	5,084 single product customers	Moving customers from 1 to 2 products drops churn from 27.71% to 7.58%
6	Review female customer experience — products, communication, and service	4,543 female customers	Female churn is 8.61% higher than male — a consistent gap that needs investigation

Priority 3 — Strategic (1 to 3 Months)

#	What to Do	Who to Target	Why
7	Build a real-time inactivity monitoring system that flags customers before they churn	All customers with declining activity	Inactivity precedes churn — catching it early allows pre-emptive action
8	Investigate the root cause of Germany underperformance — competitive landscape, service quality, pricing	Germany market team	Germany's problems span every variable. An external factor — competitor or internal failure — likely exists
9	Prioritise retention budget toward high balance customers aged 41 to 60	High and Premium balance segments	These customers carry the highest average balance. Their churn costs the most per departure
10	Design age-appropriate product bundles for the 41 to 60 segment with suitable fee structures	3,117 customers aged 41-60	The highest churn age group needs products and pricing that match their life stage and financial needs

6. Final Summary

The table below consolidates all key findings from the analysis.

Finding	Detail
Overall churn rate	20.37% — roughly double the industry benchmark
Highest churn country	Germany at 32.44% — double France (16.15%) and Spain (16.67%)
Highest churn age group	51 to 60 at 56.21% — more than half of this group left
Product sweet spot	2 products at 7.58% churn — cross-selling beyond this backfires
Most dangerous segment	3 to 4 products at 82-100% churn — aggressive over-selling
Engagement gap	Inactive members churn at 26.85% vs 14.27% for active
Gender gap	Female 25.07% vs Male 16.46% — 8.61% difference
Balance finding	Churned customers hold 18,363 more on average than retained
Funnel 1 max risk	Germany + Inactive + 41-60 + 3-4 Products = 94.87% churn (4.7x overall)
Funnel 2 max risk	Inactive + 41-60 + 3-4 Products (global) = 95.28% churn (4.7x overall)
Variables with no impact	Credit score, tenure, salary, and credit card ownership — all negligible